FUSING SELF-SUPERVISED AND SUPERVISED LEARNING: A UNIFIED PRE-TRAINING FRAMEWORK FOR CROP MAPPING

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***Abstract*—Crop identification through automated approaches is vital for tracking agricultural activities and ensuring food security, making it a key interest for the Earth Observation (EO) community. However, the scarcity of labeled data in diverse regions and the limitations of existing methods pose significant challenges. This study addresses these issues by proposing a unified framework that integrates supervised and self-supervised learning within the pre-training process. Our approach leverages ESA Sentinel-2 satellite time-series data, combining spatio-spectral and temporal features with labeled Land Use Land Cover (LULC) data as an auxiliary supervised task. Through this dual-task framework, we extract robust, generalizable features for crop identification, even in data-scarce settings. Moreover, while the dual model demonstrated superior performance, the addition of derived vegetation indices did not significantly impact its effectiveness, highlighting the strength of the features learned directly by the deep learning model.**

***Index Terms*—Sentinel-2 time series, Transformers, Self-supervised Learning, Supervised Learning, Few-shot Learning, Vegetation Indices.**

I. INTRODUCTION

Automated crop identification is essential for tracking on-farm agricultural activities and for ensuring the food security of a region, as it enables the precise assessment of the crop extent and production. The availability of satellite data, particularly Sentinel-2 Earth Observation (EO) time series, has revolutionized this process by providing high-resolution spatio-temporal data capturing plant phenology, enhancing crop and land-cover classification. Global initiatives, such as the European Space Agency's (ESA) Land Use Land Classification (LULC) mapping and Google's Dynamic World Cover layers, further improve vegetation and land cover mapping with labeled datasets, making remote sensing a cornerstone of modern agricultural analysis.

Despite these advancements, significant challenges remain in using remote sensing data for crop classification. Labeled data is still scarce in many regions, limiting the development of robust models that can generalize across diverse geographies. Traditional approaches, such as using handcrafted features (e.g. spectral statistics, phenological parameters, curve fitting coefficients, and others) [1] [2] [3], have demonstrated limitations due to their sensitivity to noise and dependence on specific smoothing techniques. While deep learning methods like CNNs [4], RNNs [5], and Transformers [6] [7] have shown improved performance, these models often require extensive labeled datasets and are prone to overfitting in data-scarce settings. Techniques such as METEOR [8] and Satellite Image Time Series-Former (SITS-Former) [9] have sought to address these issues by leveraging spatio-spectral and temporal information. Meta-learning Tailored for Earth Observation (METEOR) employs Meta-learning for few-shot adaptation, while SITS-Former uses self-supervised learning to harness unlabeled cropland data.

Adopting only METEOR or SITS-Former has limitations: METEOR excels in few-shot adaptation but underutilizes unlabeled data, while SITS-Former leverages self-supervised learning but may not fully exploit labeled datasets or align pre-training with specific tasks. Combining supervised auxiliary tasks with self-supervised learning in a unified pre-training framework addresses these gaps by utilizing both labeled and unlabeled data effectively, enhancing generalization, mitigating overfitting, and optimizing task-specific performance.

We propose a unified framework that combines the strengths of both supervised learning (SL) and self-supervised learning (SSL). Specifically, we aim to integrate both tasks into a unified pre-training framework. In this approach, self-supervised learning focuses on extracting generalizable features from unlabeled data, while labeled LULC data serves as an auxiliary supervised task to guide the model in learning representations that align with land-cover classification. This combination enables the model to develop robust feature extractors transferable to downstream crop classification tasks, even in data-scarce settings.

The contributions of this work are as follows,

- First, we utilize Sentinel-2 based spatio-spectral time series labeled LULC data for robust feature extraction, demonstrating its adaptability to crop classification, particularly in data-scarce settings.
- Second, we integrate SSL and SL to achieve a balance between unsupervised pre-training and fine-tuning with limited labeled data.
- Finally, contrary to the observations in the literature [10], [11], we noticed that the inclusion of different satellite derived indices did not aid in model's performance improvement.

II. METHODOLOGY

A. Data Preparation

Sentinel-2 L2A Level-2A Surface Reflectance values were obtained via the Google Earth Engine (GEE) for both pre-training and the crop classification tasks. The dataset, covering the period from January 1 to December 31, 2021,

was filtered to include only those images with a cloud coverage $\leq 10\%$.

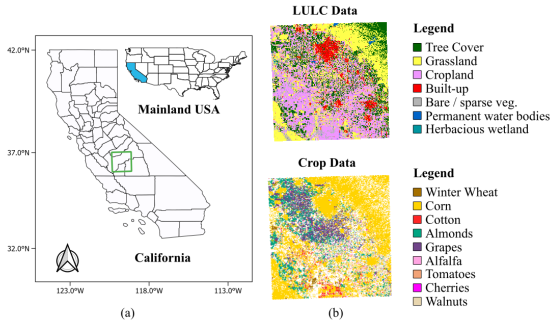


Fig. 1: (a) Location of Sentinel-2 tile used for constructing datasets. (b) Location of study area: T11SKA in California. The LULC and Crop Data images of study area are 2021 ESA WorldCover and Crop Data Layer (CDL) maps respectively.

1) Pre-Training Task Data: The labeled LULC dataset for pre-training was derived from the ESA WorldCover, a globally consistent LULC map featuring 10 classes based on Sentinel-2 imagery at 10-meter resolution. For this study, we collected $18,705$ samples of time-series data from the T11SKA region in California (see Figure 1), aligned with the ESA WorldCover dataset, which includes 7 land-use/land-cover classes: Trees, Shrubland, Grassland, Cropland, Built-up, Herbaceous Wetlands, and Bare/Sparse Vegetation. The samples were extracted at 15-day intervals using 10 spectral bands (B02, B03, B04, B05, B06, B07, B08, B8A, B11, and B12) and a consistent 5×5 patch size.

2) Crop Classification Task Data: The labeled dataset for crop classification was derived from the Cropland Data Layer (CDL) and its corresponding confidence layer, an annually updated, high-accuracy crop-specific land cover map produced by the USDA that integrates field survey data and 30-meter resolution satellite imagery to classify over 100 land cover and crop categories. For this study, nine crops—winter wheat, corn, cotton, grapes, almonds, alfalfa, tomatoes, cherries, and walnuts—were selected based on their diverse phenological cycles and cultivation practices in the T11SKA region of California, ensuring robust model evaluation across heterogeneous agricultural landscapes. The data used for crop classification was similar to the pre-training task, with the key difference being that the ground truth was derived from CDL labels with a confidence level $>$80%. The training dataset included 5 samples per class (45 total), while the test dataset comprised 89,980 samples.

B. Architecture

Our framework comprises three modules: an image patch embedding module, a Transformer encoder module, and a task module (pre-task during the pre-training phase), as illustrated in Fig. 2. The image patch embedding module processes each patch independently, extracting spatio-spectral features and converting the input time series into a sequence of embedding vectors. These embeddings are then fed into the Transformer encoder module, which incorporates temporal context by facilitating interactions across time. The resulting enriched temporal-context representations are subsequently utilized in downstream tasks.

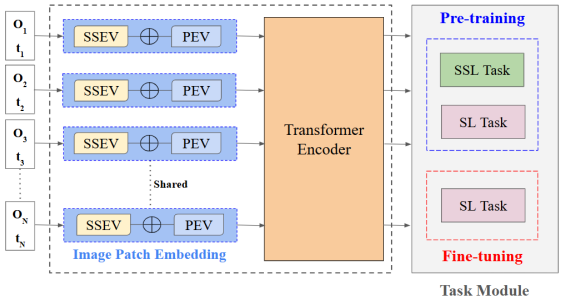


Fig. 2: Dual SITS-Former comprises three modules: an image patch embedding module (blue) shared across timesteps, a Transformer encoder (orange), and a task module (gray), enabling pretraining with SSL and SL tasks and fine-tuning for downstream tasks.

1) Image Patch Embedding Module: The image patch embedding module converts each input tuple, $\langle O_i, t_i \rangle$ (where O_i is a 3D tensor of size $10 \times 5 \times 5$ and t_i, the acquisition time, Day of Year) into an embedding vector comprising two components: a learnable spatio-spectral embedding vector (SSEV) and a predefined positional encoding vector (PEV). SSEV is generated through a lightweight network featuring two Conv3D layers with ReLU activations and batch normalization, ensuring stability and computational efficiency. Meanwhile, PEV encodes the acquisition time of an image as Day Of Year, enabling the network to handle irregular time series data effectively. This study utilizes sinusoidal positional encoding [12] to compute PEV, ensuring time-related information is encoded without introducing additional parameters.

2) Transformer Encoder: The Transformer encoder module consists of multiple layers of multi-head self-attention mechanisms, which use scaled dot-product attention to incorporate temporal information into the patch representations. Each attention head computes interactions between queries, keys, and values derived from the embedding vectors, enabling the model to learn temporal dependencies. The resulting enriched representations are then passed down to the downstream task(s). Thus, image patch embedding and the transformer encoder serve as the core feature extraction backbone that is shared across the pre-training and fine-tuning phases.

3) Task Module: The task module differs between the pre-training and fine-tuning phases. During pre-training, a dual-task approach is employed, combining LULC classification as a supervised auxiliary task with a self-supervised learning (SSL) task. In contrast, the fine-tuning phase focuses solely on the target crop classification task. These phases are discussed in detail in the following sections.

C. Pre-training stage

Some studies [8] suggest using LULC classification as a pre-training task, while others advocate for self-

supervised learning [9] as an alternative pre-training strategy. Building on these insights, our framework combines these two tasks—LULC classification and a self-supervised task—into a unified pre-training framework to develop a rich feature extraction backbone capable for learning complementary features from both the tasks.

Classification Head: The classification head consists of a max-pooling operation followed by a linear classification layer. Given a labeled LULC dataset of m sample pairs, each sample pair contains $x^{(s)}$ - the reflectance time series and $y^{(s)}$ the one hot encoding of its class label, the classification head maps extracted features to class probabilities, and the classification loss is calculated using the cross-entropy loss $\mathcal{L}_{\text{C}} = CE(y^{(s)}, \hat{y}^{(s)})$, where $\hat{y}^{(s)}$ is the predicted class distribution.

Self-Supervision Head: The self-supervision head predicts masked data using a linear layer that outputs values based on the temporal context of the input. The temporal prediction task, inspired by [9], trains the network to reconstruct missing portions of an input time series. For each sample, 30% of the patches are randomly masked and replaced with a shared padding matrix (mask), containing random values drawn from a normal distribution. The network learns to recover these masked patches using their spatiotemporal context and acquisition dates, enabling it to capture meaningful relationships.

The loss for this task minimizes the L_2 distance between the predicted and true reflectance values at the masked timesteps. It is defined as: $\mathcal{L}_{\text{S}} = \sum_{s=1}^{m} \left\| M^{(s)} \odot \left(v^{(s)} - W_p T^{(s)} \right) \right\|_2^2$, where $T^{(s)}$ and $v^{(s)}$ are predicted and true reflectance values, W_p is a learnable weight matrix (linear layer), $M^{(s)}$ is a binary mask (1 for masked timesteps, 0 otherwise), $\odot$ denotes the element-wise product, and $\| \cdot \|_2$ is the L_2 norm.

The total loss for pre-training task is a weighted combination of the supervised LULC classification task loss ($\mathcal{L}_C$) and the self-supervised temporal prediction loss ($\mathcal{L}_S$). Let w_0 and w_1 be the weights for these losses respectively. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = w_0 \mathcal{L}_{\text{C}} + w_1 \mathcal{L}_{\text{S}}$$

During training, the model is optimized to minimize $\mathcal{L}_{\text{total}}$. By balancing the weights w_0 and w_1, the model effectively learns to incorporate temporal dynamics and spatial-spectral characteristics for effective feature representations.

Data augmentation helps networks learn invariant features and reduce overfitting. We use two augmentation techniques: flipping and rotation, with the same technique applied to all patches in a time series.

D. Fine-tuning stage

After pre-training, a classification module is added to the Transformer Encoder framework, as illustrated in Fig. 3. The module contains a max-pooling layer followed by a linear fully connected (FC) layer. The entire model (including the feature extraction backbone with the pre-trained weights) is then fine-tuned end-to-end on a small set of labeled crop data in a few-shot learning context.

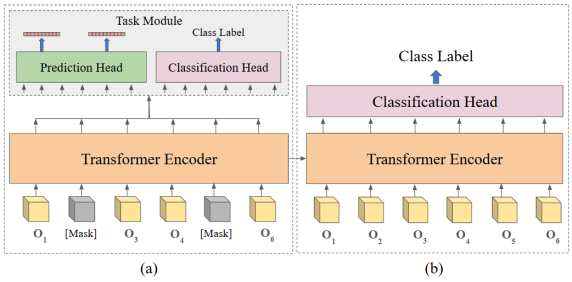


Fig. 3: (a) Pre-training stage. (b) Fine-tuning stage. In the pre-training stage, general features are learned from LULC Sentinel-2 time series through SSL proxy task and SL classification, while in the fine-tuning stage, the model is adapted for crop classification in data-scarce regions with minimal labeled data.

III. Results

A. Experimental setup and Evaluation metrics

In our experiments, we evaluated the performance of Dual SITS-Former against four other learning techniques on a k-shot crop classification task. Each model employs a distinct learning strategy during pre-training. Dual SITS-Former integrates self-supervised learning (SSL) and supervised learning (SL) for feature extraction, while SITS-Former relies solely on SSL, the Transfer Learning approach uses SL, and the Meta-learning models are pre-trained on LULC-labeled data to enable rapid adaptation to new tasks with minimal labeled examples.

The Dual SITS-Former is pre-trained on LULC data for 200 epochs, followed by fine-tuning on k-shot crop classification data for 100 epochs. During pre-training, $\mathcal{L}_{\text{S}}$ and $\mathcal{L}_{\text{C}}$ were weighted at 0.95 and 0.05, respectively, to balance the loss scales. Setting these weights to 1 and 0, respectively, transforms it into a pure SSL pre-training model, SITS-Former. Conversely, setting the weights to 0 and 1 results in a pure SL pre-training model, akin to the Transfer Learning technique. Both these models were pre-trained under the same experimental settings as Dual SITS-Former. The Transformer encoder in Dual SITS-Former was configured with a hidden size of 256, three layers, and eight attention heads, enabling it to effectively capture temporal dependencies within the data.

Optimization was performed using the Adam optimizer with a batch size of 512. The initial learning rate was set to $1\text{e}{-4}$, with a warm-up phase over the first 10 epochs, after which the learning rate decayed by a factor of 0.99 per epoch. To mitigate overfitting, dropout was applied to all layers with a dropout rate of 0.1.

The Meta Learning model uses a feature extraction architecture identical to Dual SITS-Former. Meta tasks are formed using a 4-way 5-shot support set and a 4-way 5-shot query set. The Meta model is pre-trained using second-order MAML algorithm [13] with: learning rate initialized at $2\text{e}{-4}$ and dynamically adjusted with a decay factor of 0.1.

In all experiments, data augmentation techniques were applied to the pre-training datasets, with 3% of the data

reserved as the validation set. During the fine-tuning stage, all pre-trained models were fine-tuned on k-shot crop classification data for 100 epochs using a learning rate of $1e{-}4$ and a batch size of 5. To reduce overfitting, dropout with a rate of 0.1 was applied to all layers. The Scratch Model, by contrast, was trained from randomly initialized weights on the crop classification dataset for 100 epochs.

The models were evaluated on Overall Accuracy (OA), F1-score, and the Kappa coefficient. OA quantifies the overall classification quality, the F1-score evaluates the accuracy for individual classes, and the Kappa coefficient assesses classification consistency. The reported results are averaged over 50 k-shot crop classification tasks to ensure robustness and reliability tested on 89,980 samples.

B. Comparison with other techniques

The performance of various models in the 5-shot crop classification task is summarized in Table I. Dual SITS-Former achieved the highest Overall Accuracy (OA) of 73.2% and F1-score of 72.7%, outperforming the other techniques. This highlights the effectiveness of combining self-supervised learning (SSL) and supervised learning (SL) techniques for feature extraction. SITS-Former, leveraging only SSL, achieved a slightly lower performance, with an OA of 72.56% and an F1-score of 72.07%, reflecting its strength in temporal dependency modeling. The Meta Learning and Scratch models yielded lower results, achieving an OA of approximately 68.86% and an F1-score of 68.47%, indicating the challenges of training models from scratch or solely relying on Meta-learning in limited-shot scenarios.

TABLE I: 5-shot setting performance metrics of different techniques.

Model	OA (%)	F1-score (%)	Kappa
Dual SITS-Former	**73.20 ± 2.87**	**72.70 ± 2.97**	**0.698 ± 0.032**
SITS-Former	72.56 ± 2.73	72.07 ± 2.85	0.691 ± 0.031
Transfer Learning	71.64 ± 2.93	71.24 ± 3.02	0.681 ± 0.033
Meta Learning	68.862 ± 3.319	68.475 ± 3.415	0.650 ± 0.0373
Scratch	68.864 ± 3.322	68.479 ± 3.422	0.650 ± 0.0374

The confusion matrix in Fig. 4 illustrates the high classification accuracy of Dual SITS-Former across crop types, particularly excelling in crops such as Winter Wheat, Tomatoes and Corn.

C. Impact of increasing number of shots

As shown in Table II, increasing the number of labeled samples (k-shots) significantly improved the OA for both Dual SITS-Former and SITS-Former models. Dual SITS-Former achieved the highest OA at 78.63% for 15-shots, consistently outperforming SITS-Former across all shot levels with a notable improvement of approx. 1% in OA at 10-shots. While SITS-Former showed competitive performance, our approach demonstrated superior adaptability to limited labeled data, indicating its effectiveness in few-shot learning scenarios.



Fig. 4: Dual SITS-Former

TABLE II: Overall Accuracy (%) over the k-shots

Model	1-shot	5-shot	10-shot	15-shot
Dual SITS-Former	**43.16 ± 6.15**	**73.2 ± 2.87**	**78.56 ± 2.15**	**78.63 ± 1.79**
SITS-Former	**43.16 ± 6.15**	72.56 ± 2.73	77.48 ± 1.92	78.6 ± 1.79

D. Impact of Vegetation Indices

Prior literature [10], [11] demonstrates that incorporating vegetation indices (VI) can significantly enhance crop classification performance, with models trained on specific indices often achieving comparable or superior results to those trained on all spectral bands. Building on this insight, our experiments utilized a set of indices—NDVI, n-NDVI, NDRE, NDRE2, NDRE3, NDMI, and ARI—to effectively capture crop phenological variations [10], [11].

Contrary to observations in the literature, we found that incorporating VIs alongside spectral bands led to a reduction in performance for most models, as shown in Fig. 5. Dual SITS-Former maintained relatively stable performance, with only a marginal decrease in OA. In contrast, SITS-Former was the only model to exhibit an improvement in the OA. These results call into question the need for human-crafted vegetation indices and highlight the potential usefulness of features automatically learned by deep learning models.

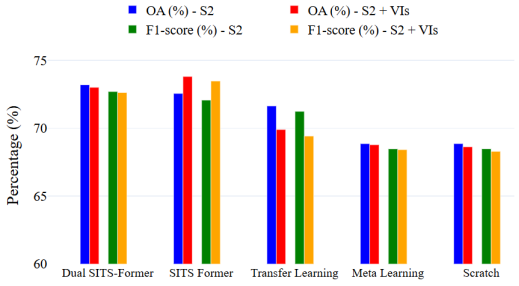


Fig. 5: Performance comparison with and without Vegetation Indices.

IV. Conclusion

We proposed a unified pre-training framework integrating supervised and self-supervised learning for remote-sensed time series classification, achieving improved crop classification performance in few-shot learning scenarios, evaluated in California. Incorporating supervised learning (SL) tasks during pre-training enhanced feature extraction by aligning representations with labeled LULC data, enabling robust crop classification in data-scarce regions.

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